# Akshat Neema

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## Professional Summary

Computer Science and Engineering student at PDPM IIITDM Jabalpur (GPA: 8.1/10.0) with hands-on experience building full-stack applications, scalable mobile apps, RESTful APIs, and deep learning systems. Proficient in DSA, OOP, DBMS, and Operating Systems. Active open-source contributor with agile, team-based development experience. Seeking an SDE role to build reliable, high-performance systems.

## Education

**PDPM IIITDM Jabalpur** — Expected May 2027

*B.Tech in Computer Science and Engineering — GPA:* ***8.1 / 10.0*** — *Jabalpur, Madhya Pradesh*

1. Relevant Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP, System Design

## Experience

**PDPM IIITDM Jabalpur — Fusion ERP (Open-Source)** — Aug 2025 – May 2026

*Backend Developer, Hostel Management Module | Python, Django, PostgreSQL, Git* — *Jabalpur, MP*

- Engineered backend features for an institute-wide ERP system serving **1,000+ students**, directly elevating room allocation accuracy and hostel complaint resolution workflows end-to-end.
- Reduced manual administrative overhead by **40%** by designing automated room allocation and complaint tracking pipelines, replacing error-prone manual processes with auditable workflows.
- Collaborated in a distributed open-source team via Git — submitted pull requests, conducted code reviews, and resolved issues following production-grade contribution standards.
- Authored and executed unit tests for critical backend logic, ensuring zero regression defects across sprint cycles and strengthening overall system reliability.

## Projects

**AI-Based Osteoporosis Detection System** | *Python, PyTorch, ResNet, Grad-CAM* — 2024

- Engineered an end-to-end deep learning pipeline using ResNet, achieving **93% validation accuracy** with <**5% false-negative rate** on knee X-ray osteoporosis classification.
- Applied transfer learning and automated data augmentation, boosting dataset quality by **35%** and cutting training time by **40%** compared to training from scratch.
- Integrated Grad-CAM explainability to generate clinician-readable visual heatmaps, surfacing model decision regions and reducing black-box behavior.

**Salon Booking Platform** | *Flutter, Firebase, Firestore* — 2025

- Architected a cross-platform mobile app (iOS & Android) enabling real-time salon booking and management for multiple user roles.
- Implemented role-based access control (RBAC) with OTP authentication and a Firestore NoSQL schema using atomic transactions to eliminate concurrent booking conflicts.
- Modeled a real-time NoSQL database schema using Firebase Firestore to handle concurrent bookings, reducing appointment conflicts by enforcing atomic transaction writes.
- **Demo:** Watch the full app walkthrough on YouTube — booking flow, RBAC, and real-time updates in action.

## Technical Skills

**Languages:** C++, Java, Python, Dart, Kotlin, C

**Web & Backend:** REST APIs, HTML, CSS, JavaScript, React.js, Django

**Frameworks & Libraries:** Flutter, TensorFlow, Keras, PyTorch

**Databases:** MySQL, MongoDB, Firebase Firestore, PostgreSQL

**DevOps & Tools:** Git, GitHub, CI/CD, VS Code, Android Studio, Jupyter Notebook

**Core CS Concepts:** OOP, Operating Systems, DBMS, Computer Networks, System Design, DSA

## Achievements & Certifications

1. Selected for **Adobe GenSolve Hackathon**, competing against teams from premier engineering institutes across India.
2. Active open-source contributor — submitted pull requests for feature development and bug fixes on production codebases.
3. Completed a minimum of **100 hours** dedicated to algorithmic practice on platforms like **LeetCode** and **Codeforces**, leading to mastery of key concepts and establishing a strong foundation for future software engineering projects.
4. **HP LIFE** — HP Power Lab Program (Entrepreneurship & Technology)
5. **Pod** — Personal Branding & Professional Development Certification
6. **Adobe GenSolve** — Earned recognition for competitive performance in a national-level hackathon; certified participant among top engineering teams across India.